

When does compact differentiation beat pseudospectral?

Cutoff energy, cell Reynolds number and a dealiased control in two-dimensional decaying turbulence

Leonardo Araujo

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Abstract

San and Staples (2012) compared high-order finite differences with a Fourier–Galerkin method in decaying two-dimensional turbulence. They found sixth-order schemes in agreement with the spectral method at a fraction of its cost, and organized resolution by the cell Reynolds number Re_c . We revisit that comparison with four controls. Explicit sixth-order, Lele’s compact sixth-order and pseudospectral differentiation, the last undealiased, with the $2/3$ rule and with $3/2$ padding, are applied through their symbols in Fourier space by one solver, so that only the differentiation changes. Accuracy is the resolved spectral bandwidth against 2048^2 explicit sixth-order references, cross-validated against compact 2048^2 runs for the most demanding cases. Cost at equal resolved range is estimated from measured GPU step times, the advective time-step limit of each scheme, and the measured growth of the explicit scheme’s resolved range with the grid. In five decaying cases on two grids, cases with the same Re_c fall in every regime, while the reference’s energy at the grid’s Nyquist shell relative to its peak, χ , largely organizes the ordering.

- For $\chi \lesssim 10^{-8}$ pseudospectral differentiation resolves the most, up to 1.7 times the shells of explicit sixth order at 0.54–0.76 times its cost at equal resolved range.
- Between about 10^{-7} and 2×10^{-6} pseudospectral and compact sixth order trade places; near 5×10^{-6} the compact scheme leads clearly. Wherever $\chi \lesssim 5 \times 10^{-6}$ it resolves 1.3–1.6 times the shells of explicit sixth order at 0.42–0.78 times its cost.
- For $\chi \gtrsim 10^{-5}$ neither beats explicit sixth order.

Pseudospectral differentiation piles energy up at the cutoff, and $3/2$ padding leaves both the pile-up and the resolved bandwidth unchanged (within 8 shells in all ten cases) at 1.85 times the step cost. So the crossover with compact differentiation is not an artifact of quadratic aliasing of the nonlinear term. In forced, statistically stationary turbulence the schemes cannot be told apart in the inertial range within the sampling scatter of the forcing.

1 Introduction

Finite differences of high order are attractive for turbulence simulation because they combine accuracy at the resolved scales with the flexibility of local operators. Lele’s compact schemes [1] are the standard example. A tridiagonal implicit relation between derivatives and values gives a modified wavenumber close to the exact one over most of the resolved range, so they are described as having spectral-like resolution. Laizet and Lamballais [2] built on this a simple and efficient solver for incompressible flow with quasi-spectral accuracy. It used sixth-order compact schemes, a modified-wavenumber analysis of every operator, and a Poisson equation solved in spectral space. Through a series of validation cases they showed that, at marginal resolution, the compact schemes preserve quantities sensitive to the small scales much better than second-order schemes, for a moderate extra cost. They also argued that compact differentiation of the nonlinear terms is cheaper than pseudospectral differentiation. The resulting code, Incompact3d [3, 4], scales to $O(10^5)$ cores, and its successor Xcompact3d [5] is a widely used open-source framework. The quasi-spectral philosophy is therefore well established.

The comparison with spectral methods at the resolution limit is more delicate. Ghosal [6] and Kravchenko and Moin [7] analysed the numerical errors of large-eddy simulation. They showed that aliasing errors are more destructive for spectral and high-order finite-difference calculations than for low-order ones, and that the form of the nonlinear term decides whether a scheme conserves energy in the presence of aliasing. Moin and Mahesh [8] summarise the resulting picture: finite-difference schemes carry larger differentiation errors but lower aliasing errors, and the balance of the two sets the error at the small scales. Fedoun, Lardjane and Gökalp [13] separated aliasing from truncation errors in physical and spectral space, and found high-resolution compact differences in the convective form to perform nearly as well as dealiased pseudospectral differentiation in large-eddy simulation of decaying isotropic turbulence. Park, Yoo and Choi [14] found the finite-differencing error to outweigh the aliasing error in large-eddy simulation of isotropic turbulence, with the skew-symmetric form reducing both. Truncation errors alone can distort the cascade near the cutoff: Yalla, Oliver and Moser [15] traced an energy pile-up in implicitly filtered large-eddy simulation to numerical dispersion, without any aliasing.

The closest work to ours is that of San and Staples [9]. They compared explicit central differences of orders 2–6, compact schemes of orders 4 and 6, Arakawa and dispersion-relation-preserving schemes with a Fourier–Galerkin method in decaying two-dimensional turbulence, in the vorticity–streamfunction form with a fast Poisson solver shared by all schemes. Sixth-order schemes agreed with the spectral method, at speed-ups of about 5 in CPU time. They organized the resolution by the cell Reynolds number $Re_c = Re h$ ($Re 2\pi/N$ on their domain). Below $Re_c \approx 2$ all schemes agreed; for larger Re_c the order mattered; and above $Re_c \approx 30$ the non-conservative schemes piled energy up at small scales, which they attributed to aliasing and which the conservative Arakawa scheme removed. Their spectral reference was dealiased by padding to $2N$ points, and its reported cost interpolated between the times on N and $2N$ points. Recently, Rodhiya, Bhattacharya and Verma [10] found that forced three-dimensional turbulence computed with a pseudospectral and a second-order finite-difference solver has nearly the same statistics, and Yamamoto and Tsuji [17] found that in channel-flow simulation the skew-symmetric form with a selective filter, not full $3/2$ dealiasing, gives the best accuracy at equal cost.

Three questions are left open. Is Re_c , a property of the grid and the viscosity, the right axis, or does the answer depend on how much of the flow’s spectrum actually reaches the cutoff? Is the high-wavenumber pile-up aliasing, as is usually assumed, or truncation? And which scheme gives the most resolved bandwidth per unit of computing time? Answering them needs four things together.

1. The schemes compared in otherwise identical solvers, so that differences come from the differentiation alone.
2. Pseudospectral differentiation as a competitor rather than as the reference: undealiased, with the $2/3$ rule, and with $3/2$ padding, so that aliasing can be switched off without giving up range.
3. A reference whose own discretization does not affect the measures, checked by a second reference computed with a different scheme.
4. A cost that includes the time step each scheme allows, not only the time per step, with each scheme in its fastest implementation.

This paper provides such a comparison for two-dimensional periodic turbulence, in an open code written for the purpose (cnavier). Its contributions are:

- an experimental apparatus in which explicit sixth-order, compact sixth-order and pseudospectral differentiation are applied through their symbols in Fourier space by the same solver, verified to reproduce the sparse finite-difference operators to round-off (sections 2 and 3);

- resolved bandwidth against cross-validated high-resolution references as the accuracy measure, and an estimate of the cost at equal resolved range from measured step times and the advective stability limit (section 4);
- a regime map for decaying turbulence: the energy at the grid cutoff relative to the peak, χ , largely organizes which scheme resolves most and costs least, from pseudospectral through compact to explicit sixth order, with a crossover band between pseudospectral and compact, while cases at equal Re_c span all regimes (section 5.1);
- a dealiased control: 3/2 padding leaves the pseudospectral pile-up and bandwidth unchanged, so quadratic aliasing of the nonlinear term does not explain the crossover (section 5.2);
- the finding that this organization does not carry over to forced, statistically stationary turbulence, where the schemes are indistinguishable in the inertial range within the sampling scatter of the forcing (section 5.4).

We do not propose a new scheme. The schemes are standard; what is new is the controlled comparison and the regime-dependent answer.

2 Numerical method

2.1 Equations and time integration

The two-dimensional incompressible Navier–Stokes equations are solved on the doubly periodic unit square in the vorticity–streamfunction form

$$\frac{\partial \omega}{\partial t} = N(\omega) + \nu \nabla^2 \omega + f, \quad \nabla^2 \psi = -\omega, \quad u = \partial_y \psi, \quad v = -\partial_x \psi, \quad (1)$$

with $\nu = 1/\text{Re}$ and f the forcing and damping terms of section 5.4. The nonlinear term takes the skew-symmetric form

$$N = -\frac{1}{2} (u D_x \omega + v D_y \omega) - \frac{1}{2} (D_x(u\omega) + D_y(v\omega)). \quad (2)$$

Because the periodic difference operators are antisymmetric, it conserves the discrete enstrophy $\frac{1}{2} \langle \omega^2 \rangle$ to round-off for every scheme here. The advective form does not, and in under-resolved turbulence creates enstrophy near the cutoff, 23–45 % of the input in the runs of the companion methodology. Time integration is the classical fourth-order Runge–Kutta method, in double precision throughout.

2.2 Operators by their symbols

On a periodic grid of n points every translation-invariant difference operator is a circulant matrix, diagonal in the discrete Fourier basis. Its eigenvalue at wavenumber k is its symbol, $\sum_j c_j e^{ikjh}$ for the stencil c_j . Writing $\tilde{D}_x(k) = ik^*(k)$ and $\tilde{D}_{xx}(k) = -(k^{**})^2$, a scheme is characterised by its modified wavenumbers. The solver reads the symbols from the first row of each one-dimensional operator and keeps them in one table. Everything that depends on the operators uses that table: the Poisson eigenvalues $\tilde{D}_{xx} + \tilde{D}_{yy}$, the energy of a mode in the spectral diagnostics, the amplitude of the random forcing, and the stability limits.

Every operator is applied in Fourier space: one forward transform of ω , then inverse transforms of the symbols times its spectrum for $D_x \omega$, $D_y \omega$, the Laplacian, the stream function and the velocity. The skew-symmetric correction takes two more forward transforms and one inverse. For a finite-difference scheme this is the same matrix as the sparse stencil, applied differently: after 20 steps of a forced run with every term active, the two paths agree to 10^{-14} . Changing the scheme changes only the symbol table. This is what makes the comparison controlled. The time integrator, the Poisson operator (always $\tilde{D}_{xx} + \tilde{D}_{yy}$ of the scheme itself), the nonlinear form, the forcing and the diagnostics are identical code for all schemes.

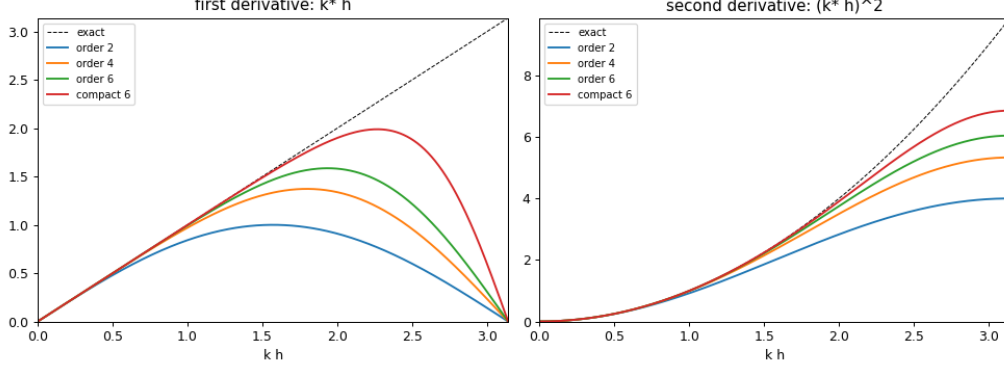


Figure 1: Modified wavenumbers of the first and second derivatives: explicit orders 2, 4 and 6, and compact sixth order. The pseudospectral symbols are the exact ones (dashed).

fraction of $[0, \pi]$ within	first derivative		second derivative		$(k^*h)_{\max}$
	1 %	10 %	1 %	10 %	
E6	0.35	0.54	0.45	0.71	1.59
C6	0.50	0.70	0.56	0.81	1.99
S	1	1	1	1	π
$S_{3/2}$	1	1	1	1	π
$S_{2/3}$	2/3	2/3	2/3	2/3	$2\pi/3$

Table 1: Fraction of the resolved range differentiated within 1 % and 10 %, and the largest modified wavenumber, which sets the advective stability limit.

2.3 The schemes

Explicit sixth order (E6). The centred seven-point stencils.

Compact sixth order (C6). Lele’s tridiagonal schemes [1], $\frac{1}{3}f'_{i-1} + f'_i + \frac{1}{3}f'_{i+1} = \frac{14}{9}\delta_1 f_i + \frac{1}{9}\delta_2 f_i$ and the analogous second-derivative scheme with $\alpha = 2/11$. On a periodic grid $A^{-1}B$ is circulant; its symbols are $k^*h = (\frac{14}{9}\sin\theta + \frac{1}{18}\sin 2\theta)/(1 + \frac{2}{3}\cos\theta)$ and $(k^{**}h)^2 = -(\frac{24}{11}(\cos\theta - 1) + \frac{3}{22}(\cos 2\theta - 1))/(1 + \frac{4}{11}\cos\theta)$, $\theta = kh$.

Pseudospectral (S). The exact symbols ik and $-k^2$, with the first-derivative symbol set to zero at the Nyquist wavenumber. With exact symbols the two discrete energies $\frac{1}{2}\langle |\mathbf{u}|^2 \rangle$ and $\frac{1}{2}\langle \psi\omega \rangle$ coincide.

Pseudospectral with the 2/3 rule ($S_{2/3}$). Orszag’s truncation [11]: modes with $|k| \geq n/3$ along either axis are removed from every operator and from ω after each step. The nonlinear term is then a Galerkin truncation and conserves energy and enstrophy exactly in either form (10^{-14} of the largest shell transfer in our tests).

Padded pseudospectral ($S_{3/2}$). The exact symbols, with the nonlinear term dealiased by padding [12]: the spectra of u , v , $D_x\omega$ and $D_y\omega$ are zero-padded to a grid 3/2 as fine, the product is formed there and its spectrum truncated back. Every mode but the Nyquist ones then carries the Fourier–Galerkin nonlinear term, free of quadratic aliasing, and no retained non-Nyquist mode is discarded. This is the dealiased pseudospectral baseline of the comparison; $S_{2/3}$ buys the same freedom from aliasing by giving up a third of the range.

Figure 1 and table 1 summarise the modified wavenumbers. The larger the resolved fraction, the larger $(k^*h)_{\max}$ and the smaller the explicit advective time step, which goes as $h/(k^*h)_{\max}$.

2.4 Spectral diagnostics

The diagnostics use the solver’s own operators. The energy of mode $\mathbf{k}$ is $\frac{1}{2}A|\hat{\psi}|^2$ with $A = |\tilde{D}_x|^2 + |\tilde{D}_y|^2$ and $\hat{\omega} = Q\hat{\psi}$, $Q = -(\tilde{D}_{xx} + \tilde{D}_{yy})$. The spectral energy and enstrophy fluxes use the solver’s nonlinear term, weighted by A/Q for the energy. With these definitions the discrete energy budget closes exactly, to round-off, rather than to the order of the scheme. The energy spectrum $E(k)$ is the shell sum of mode energies in shells of width 2π .

3 Verification

- *Order.* On a manufactured periodic solution the explicit schemes converge at orders 2, 4 and 6 and the compact scheme at 6.04, with an error at 64^2 twelve times smaller than E6. The pseudospectral error at 32^2 is 3.5×10^{-9} , against 1.1×10^{-4} for C6, and falls by 16 when the time step is halved: it is the error of the time integration.
- *Equivalence of the two operator paths.* For E6 and C6, the Fourier path reproduces the sparse operators to 10^{-14} in the fields, integrals and spectra after 20 forced steps. A decaying turbulence run with C6 reproduces the spectrum of the sparse (banded) implementation to 4×10^{-12} .
- *Padding.* The $S_{3/2}$ nonlinear term matches the direct convolution of the modal coefficients over all non-Nyquist triads to 7×10^{-15} on 10×8 and 9×7 grids, which covers the even and odd Nyquist cases.
- *CPU and GPU.* The GPU implementation (cuFFT) matches the CPU one (FFTW) to 6×10^{-15} for all five schemes, including both dealiased ones.
- *Conservation.* The skew-symmetric form conserves enstrophy to 10^{-15} – 10^{-14} of the largest shell transfer for all schemes, and $S_{2/3}$ and $S_{3/2}$ also energy, in either form.
- *References.* Each decaying case has its own 2048^2 E6 reference. For the two most demanding cases ($\text{Re} = 8 \times 10^4$, $k_0 = 20$), a second 2048^2 reference computed with C6 agrees with it to 1.5×10^{-3} up to $K = 255$, the largest shell compared, and to 10^{-4} up to $K = 127$. The references therefore do not depend on their own discretization within the tolerances measured, 2 % and 10 %.

4 Experimental design

Decaying turbulence. The initial field has random phases, with modal amplitudes whose shell envelope is $E(k) \propto (k/k_0)^4 e^{-2(k/k_0)^2}$ peaking at $K = k_0$ ($K = |\mathbf{k}|/2\pi$), normalised to $E = \frac{1}{2}$. The phase of each wavevector depends on the seed and the wavevector only, so every grid that holds the spectrum gets the same field. All runs of a case share the time step (6.25×10^{-5}), so the spectra are compared at exactly the same instant. The comparison time keeps t/τ_0 fixed, with $\tau_0 \propto 1/k_0$ since all runs start with the same energy: $t = 0.09 \cdot 10/k_0$, about six eddy turnovers. By then the cascade has filled the spectrum, but the largest scales ($K \leq 20$) still agree with the reference to 10^{-3} – 10^{-2} , so the runs have not yet decorrelated; at $t = 0.5$ they differ by tens of per cent even at $K = 10$. The baseline is $\text{Re} = 2 \times 10^4$, $k_0 = 10$. One parameter at a time is varied: $\text{Re} = 5 \times 10^3$ and 8×10^4 , and $k_0 = 5$ and 20 . Each case is run on 256^2 and 512^2 with the five schemes.

Accuracy. The resolved bandwidth is the largest shell K_{10} (K_2) up to which every shell’s energy is within 10 % (2 %) of the reference. The pile-up is the largest ratio E/E_{ref} . The

case	grid	Re_c	χ	E6	C6	S	$S_{3/2}$	$S_{2/3}$
$\text{Re} = 5 \times 10^3$	512^2	9.8	2.8×10^{-12}	144	196	249	248	163
$k_0 = 5$	512^2	39	4.2×10^{-11}	146	197	246	246	154
baseline	512^2	39	1.5×10^{-8}	152	198	235	239	143
$k_0 = 5$	256^2	78	9.0×10^{-8}	71	105	101	109	62
$\text{Re} = 5 \times 10^3$	256^2	20	2.0×10^{-7}	74	105	120	120	70
$\text{Re} = 8 \times 10^4$	512^2	156	2.3×10^{-7}	152	217	193	200	117
$k_0 = 20$	512^2	39	1.6×10^{-6}	159	213	217	217	130
baseline	256^2	78	4.9×10^{-6}	74	118	93	93	50
$\text{Re} = 8 \times 10^4$	256^2	313	1.2×10^{-5}	80	79	75	79	46
$k_0 = 20$	256^2	78	1.6×10^{-4}	89	74	74	72	49

Table 2: Resolved bandwidth K_{10} (shells within 10 % of the reference) in decaying turbulence, ordered by the reference’s energy at the grid’s cutoff, χ , with the cell Reynolds number $\text{Re}_c = \text{Re } h$ for comparison. Nyquist shells: 127 (256^2), 255 (512^2).

organizing parameter tested is

$$\chi = \frac{E_{\text{ref}}(K_{\text{Nyq}})}{\max_K E_{\text{ref}}(K)}, \quad (3)$$

the reference’s energy at the grid’s Nyquist shell relative to its peak, a measure of how much of the flow lies at the cutoff. It is set against the cell Reynolds number of San and Staples [9], $\text{Re}_c = \text{Re } h$ with $h = 1/n$ on the unit square. All runs start with the same energy, so Re_c is comparable across our cases, though not numerically with theirs. Re_c depends on the grid and the viscosity only; χ also depends on where the flow puts its energy, through k_0 and the time.

Cost. The step time t_{step} is measured on one GPU (NVIDIA RTX 3050 laptop, double precision). At the advective limit the time step goes as $h/(k^*h)_{\text{max}}$, so the cost per unit simulated time is proportional to $t_{\text{step}}(k^*h)_{\text{max}}/h$. We compare at equal resolved range, which is an estimate, not a benchmark of each equal-range configuration. The resolved range of E6 grows linearly with n (74 shells on 256^2 , 152 on 512^2 at the baseline), and its cost per unit time as n^3 . Matching a scheme’s K_{10} with E6 therefore costs $(K_{10}/K_{10}^{\text{E6}})^3$ times E6 on the same grid. We report the scheme’s cost divided by that.

Accuracy and cost use each scheme’s own fastest path. The comparison isolates the discretization through the symbols, but the cost should be that of a sensible implementation. C6 and the pseudospectral variants are applied in Fourier space, which is how they are cheapest here. E6 is cheapest as a sparse seven-point stencil (table 3), so its runs and its cost use that path. The two E6 paths are the same matrix and agree to 10^{-14} (section 3), so the choice affects the cost only, and it favours E6: in Fourier space it would cost 1.4 times as much per step.

Forced turbulence. Forced runs decorrelate, so they are compared by time-mean statistics. A random forcing injects energy at rate $\varepsilon = 0.1$ in the shell $|K - 4| \leq 1$ by white-in-time kicks; a large-scale hypodrag $-\alpha_h \psi$ removes the inverse cascade. Two cases are run on 256^2 . (a) Hyperviscosity $-\nu_h(-\nabla^2)^4 \omega$ confines the small-scale dissipation near the cutoff (statistics over $t = 6\text{--}16$). (b) Plain viscosity at $\text{Re} = 10^5$ with no hyperviscosity, against a 768^2 pseudospectral reference ($t = 4\text{--}10$).

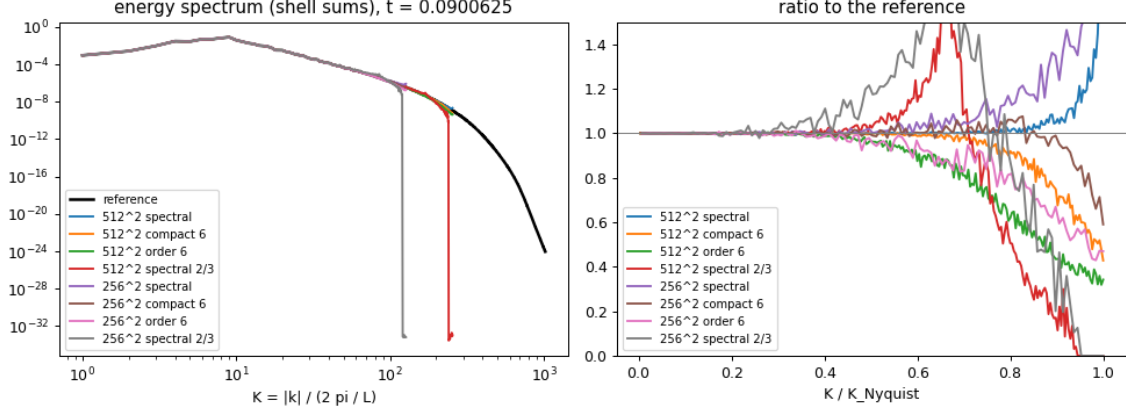


Figure 2: Baseline ($Re = 2 \times 10^4$, $k_0 = 10$) at $t = 0.09$: energy spectra on 256^2 and 512^2 and their ratio to the 2048^2 reference. The finite differences fall below the reference near the cutoff; undealiased pseudospectral piles energy up there (twice the reference at the Nyquist shell on 256^2); the 2/3 rule truncates and piles up just below its cutoff.

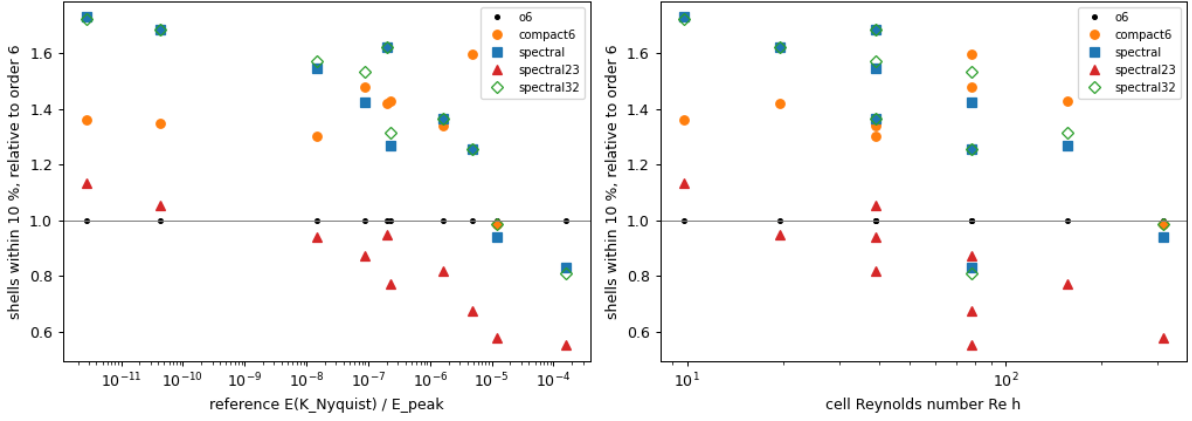


Figure 3: Resolved bandwidth relative to E6 for the five decaying cases on two grids, against χ (left) and against the cell Reynolds number $Re_c = Re h$ (right). Open diamonds: $S_{3/2}$, on top of the S squares. At $Re_c = 39$ and 78 the cases span the regimes; against χ they line up.

5 Results

5.1 Decaying turbulence: a regime map

Table 2 and fig. 3 show the result. Re_c does not organize it. The three 256^2 cases at $Re_c = 78$, differing only in k_0 , fall in three different regimes: compact and padded pseudospectral lead at $k_0 = 5$ ($\chi = 9 \times 10^{-8}$), compact clearly at $k_0 = 10$ (5×10^{-6}), explicit sixth order at $k_0 = 20$ (1.6×10^{-4}). At $Re_c = 39$, pseudospectral resolves 1.68 times the shells of E6 at $k_0 = 5$ and 1.36 at $k_0 = 20$. The case at the largest Re_c but one (156 , $Re = 8 \times 10^4$ on 512^2) has compact clearly ahead. Against χ the ordering is largely organized.

- *Resolved tail*, $\chi \lesssim 10^{-8}$: pseudospectral differentiation resolves the most, up to 1.7 times the shells of E6 and 92–98 % of the Nyquist range. The exact derivative is an advantage once the grid holds the dissipation range.
- *Crossover band*, $10^{-7} \lesssim \chi \lesssim 2 \times 10^{-6}$: pseudospectral and compact differentiation trade places, within a few shells of each other or up to 14 % apart in either order.

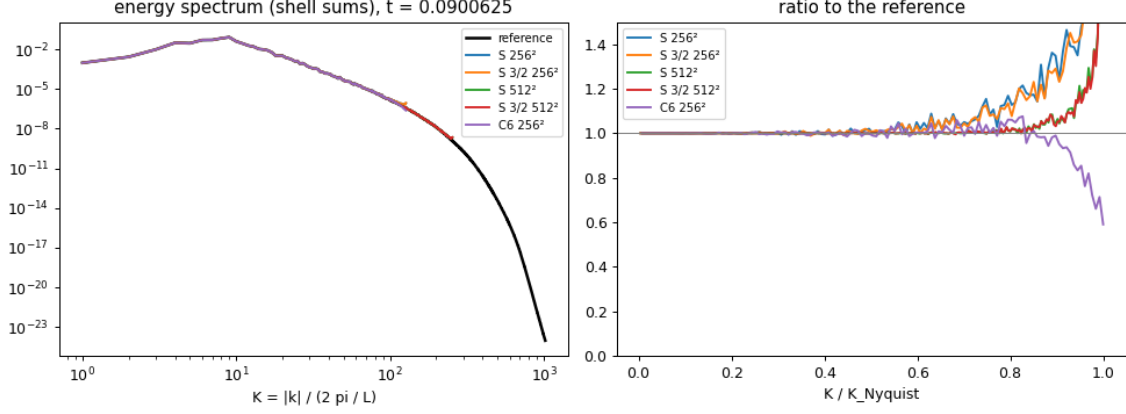


Figure 4: Baseline at $t = 0.09$: pseudospectral differentiation with and without 3/2 padding on 256^2 and 512^2 , and C6 on 256^2 , against the 2048^2 reference (right: ratio, clipped at 1.5). The padded and unpadded curves lie on top of each other.

- *Intermediate*, $\chi \approx 5 \times 10^{-6}$: the compact scheme leads clearly, with 1.6 times the shells of E6 against 1.3 for pseudospectral. Pseudospectral differentiation falls behind, dealiased or not: energy piles up at the cutoff (fig. 2; section 5.2). The finite differences underestimate the derivatives near the cutoff instead ($k^* < k$), and their spectra fall below the reference. Wherever $\chi \lesssim 5 \times 10^{-6}$, the compact scheme resolves 1.3–1.6 times the shells of E6.
- *Under-resolved*, $\chi \gtrsim 10^{-5}$: neither high-resolution scheme beats E6. Its larger differentiation error near the cutoff acts there as a numerical attenuation. This is the regime in which Kravchenko and Moin found aliasing more destructive for spectral and high-order schemes than for low-order ones [7].

The 2/3 rule caps the range at 2/3 of the Nyquist wavenumber by construction, and its spectrum piles up just below its own cutoff. It never leads; its merit is exact conservation, not resolution.

The crossover between compact and pseudospectral is not sharp. The cases at $\chi = 2.0 \times 10^{-7}$ and 2.3×10^{-7} come out in opposite order, as do those at 9.0×10^{-8} and 1.6×10^{-6} , so within the crossover band χ alone does not decide. A second parameter, perhaps the spectral slope at the cutoff, may be needed; we have not tested this.

5.2 Is the pile-up aliasing? The 3/2 control

The natural explanation of the pseudospectral pile-up is aliasing: with no numerical attenuation near the cutoff, products of resolved modes alias back onto the highest retained ones [7, 8]. $S_{3/2}$ tests it. It removes the quadratic aliasing of the nonlinear term from every mode but the Nyquist ones and keeps the full range, which $S_{2/3}$ does not.

- *Observed*. In all ten cases $S_{3/2}$ resolves the same bandwidth as S, within 8 shells (table 2; the open diamonds of fig. 3). The pile-up is the same: the largest E/E_{ref} is 1.6–2.5 with padding and without, and the two spectra are indistinguishable near the cutoff (fig. 4). Padding changes the leader in one case ($k_0 = 5$ on 256^2 , 109 shells against C6’s 105), and moves no regime boundary.
- *Established*. Quadratic aliasing of the nonlinear term is not what causes the pile-up, nor the loss of pseudospectral bandwidth relative to C6 in the intermediate regime. The crossover between compact and pseudospectral differentiation survives 3/2 dealiasing of the retained non-Nyquist modes.

ms per GPU step	E6 (sparse)	E6, C6, S (Fourier)	S _{2/3}	S _{3/2}	C6, banded sparse
256 ²	3.6	4.8	5.0	7.7	66
512 ²	12.8	17.7	18.4	31.5	269
1024 ²	51.8	78.5	82.0	–	–

Table 3: Time per step (decaying turbulence, skew-symmetric form, RK4). Applied in Fourier space, all schemes cost the same; seven-point stencils as sparse products are cheaper still. The compact operator as an exact banded circulant (78 and 45 entries per row) costs 15 times more than in Fourier space. Padding costs 1.85 times the unpadded step: five transforms on $(3n/2)^2$ points per evaluation of the nonlinear term.

- *Suggested.* The data are consistent with an accumulation caused by the truncation itself: the enstrophy the cascade carries to the cutoff is not dissipated fast enough before it, and collects in the last retained shells, as in Galerkin-truncated flows [16]. On this reading the finite differences’ $k^* < k$ near the cutoff attenuates the transfer into the last shells, and that is why their spectra fall below the reference instead.
- *Not established.* The mechanism itself. These runs rule out one explanation; testing the truncation reading, for instance with a selective filter near the cutoff, is a separate study.

5.3 Cost at equal resolved range

In Fourier space every scheme costs the same per step, about 1.4 times E6 applied as a sparse stencil (table 3). The schemes differ in their stable time step, by $(k^*h)_{\max}$, and in the range they resolve. At equal resolved range relative to E6, cost follows χ much as the bandwidth does.

- C6 costs 0.42–0.78 times as much wherever $\chi \lesssim 5 \times 10^{-6}$.
- S costs 0.54–0.76 times below about $\chi = 10^{-8}$, and 0.64–1.4 times in the crossover band and above.
- In the under-resolved cases ($\chi \gtrsim 10^{-5}$) both cost more: C6 1.9–3.1 times and S 3.4–4.9 times.
- S_{2/3} never costs less (1.3–11 times).
- Nor does S_{3/2}. It resolves what S resolves at 1.85 times the step, so it costs 1.00–1.31 times E6 for $\chi \leq 2 \times 10^{-7}$ and 2–9 times above. (Both step times from the same build: 31.5 against 17.0 ms on 512².)

The gains of the high-resolution schemes are modest, tens of per cent, where they exist. They come from resolving more shells, against a time step that is 20 % (C6) to 50 % (S) smaller than E6’s.

5.4 Forced turbulence

In forced turbulence the comparison is between time-mean statistics. With 28 forced modes, the realized enstrophy input over the averaging windows varies by about $\pm 8\%$, which sets a floor below which the schemes cannot be distinguished.

With hyperviscosity, all four schemes give the same spectral slope in the direct cascade (−3.44 to −3.48). The enstrophy flux leaves its plateau at the same shell, $K = 66$, where the hyperviscosity takes over. The plateau levels (41.9, 43.0, 46.2, 50.1 for E6, C6, S, S_{2/3}) differ as much as the total enstrophy each run removed, so they reflect the realized input, not the cascade. When the dissipation is confined near the cutoff, the differentiation scheme makes no measurable difference to the inertial range.

With plain viscosity ($\text{Re} = 10^5$, $\chi = 2.8 \times 10^{-7}$ at the 256^2 cutoff, near the decaying crossover), each spectrum is normalised by its own enstrophy flux, $E \propto \Pi_Z^{2/3}$, to remove the realized input. The spectra then agree with the 768^2 reference to 10% up to $K = 100$ (E6), 89 (S), 84 (C6) and 42 ($S_{2/3}$). The order differs from the decaying cases at the same χ , but within the remaining scatter: away from the cutoff the normalised spectra still differ from the reference by up to 8%. What stands out is how each scheme behaves at the cutoff. At $K = 120$, E6 falls below the reference (0.64) and S piles energy up (1.53, and 2.0 at the Nyquist shell). C6 lies in between (1.00, with an overshoot of up to 17% just below), and $S_{2/3}$ truncates. This agrees with the finding of Rodhiya et al. [10] that forced turbulence computed with spectral and finite-difference methods has nearly the same statistics.

6 Discussion

The results answer the three questions of the introduction for the cases tested. First, the axis. San and Staples [9] organized resolution by Re_c , a property of the grid and the viscosity. At fixed Re_c we find every regime, depending on how much of the flow's spectrum has reached the cutoff. χ measures that directly, so it is the better axis for transient decaying turbulence; its price is that it needs a reference, or an estimate of the spectrum at the cutoff. Their finding that sixth-order schemes agree with the spectral method when well resolved does not contradict ours: the resolved bandwidth is a stricter measure near the cutoff, and even at small χ pseudospectral differentiation resolves up to 1.7 times the shells of E6. Their speed-ups of about 5 compared finite differences with a dealiased spectral method whose cost was interpolated. Here one code applies every scheme, E6 as a sparse stencil and the others in Fourier space; C6 and S cost 1.4 times E6 per step, padding 1.85 times S, and the differences in cost at equal range come from these, the time step and the resolved range.

Second, the pile-up. San and Staples saw pile-up in non-conservative finite differences and attributed it to aliasing. Here the nonlinear term is skew-symmetric and conserves enstrophy for every scheme, the finite differences never pile up, and the pile-up of pseudospectral differentiation survives $3/2$ padding. Quadratic aliasing of the nonlinear term is therefore not the cause in our setting. That is consistent with a truncation effect [16], and with the dispersion-driven pile-up of Yalla et al. [15], but these runs do not identify the mechanism.

Third, the scheme. The results place the quasi-spectral argument of Laizet and Lamballais [2] on a map. Compact sixth-order differentiation is the most efficient choice over an intermediate band of resolution: where the grid is marginal but not overwhelmed. This is the regime they targeted. When the grid resolves the dissipation range, the exact symbols of pseudospectral differentiation give more bandwidth for the cost. When it is badly overwhelmed, explicit sixth order is the most robust. Its underestimated derivatives near the cutoff act as attenuation, consistent with San and Staples' observation that under-resolution calls for care in the choice of scheme [9]. The pile-up of pseudospectral differentiation at the cutoff is not the effect of quadratic aliasing: it survives $3/2$ padding unchanged (section 5.2). Both ways of dealiasing gain exact conservation; the $2/3$ rule pays for it with a third of the range, padding with 1.85 times the step, and in our measures neither pays off.

Two qualifications matter. First, χ is an organizing parameter for transient decaying turbulence before decorrelation, not a universal predictor. Near the crossover it is not sufficient on its own, and in forced statistics the inertial range does not distinguish the schemes at all. Second, the comparison measures accuracy as spectral bandwidth; other quantities, such as vortex statistics, higher-order moments or structure functions, may weigh the cutoff differently.

The apparatus has a methodological point of its own. Applying finite-difference operators through their symbols in Fourier space is natural for periodic grids, and is related to the mixed physical/spectral treatment of [2]. Here it is used as an experimental control: the scheme is a table of symbols, and the solver, the Poisson operator and the diagnostics derive from the same

table. This rules out, for instance, a Poisson operator and a transport Laplacian that disagree. It also made the compact scheme 15 times faster than its exact banded form, which the comparison of cost needed. On non-periodic or distributed grids the compact scheme needs tridiagonal solves instead, whose cost on GPUs is a subject of its own [18].

Limitations. The study is two-dimensional and periodic. It uses one time integrator (RK4), one nonlinear form (skew-symmetric) and one GPU in double precision, on which fast Fourier transforms dominate the step time. The matrix of cases is small: four perturbations of one decaying baseline and two forced cases on 256^2 , which samples Re_c at only five values. Selective filtering near the cutoff, which Yamamoto and Tsuji [17] found the most cost-effective in channel flow, was not tested. The forced comparisons are limited by the sampling scatter of the random forcing. Wall-bounded flows, where compact schemes need boundary closures and spectral methods lose their simplicity, are outside its scope.

7 Conclusions

For the tested two-dimensional periodic turbulence, no differentiation scheme is best everywhere. In decaying turbulence the ordering is largely organized by the energy at the grid cutoff, χ , and not by the cell Reynolds number: cases at equal Re_c fall in every regime. Pseudospectral differentiation resolves the most and costs the least per resolved shell when the grid resolves the high-wavenumber tail ($\chi \lesssim 10^{-8}$). Compact sixth order does so at intermediate resolution (clearly near $\chi \approx 5 \times 10^{-6}$, trading places with pseudospectral between 10^{-7} and 2×10^{-6}), and explicit sixth order is the most robust when under-resolved ($\chi \gtrsim 10^{-5}$). The crossover between compact and pseudospectral differentiation survives 3/2 dealiasing of the retained non-Nyquist modes, which leaves the pseudospectral pile-up and bandwidth unchanged at 1.85 times the step cost; it is not an artifact of quadratic aliasing. The crossovers are not sharp, and in forced, statistically stationary turbulence the schemes are indistinguishable in the inertial range within the forcing’s sampling scatter, differing only at the cutoff. The open question we leave is whether a second parameter, such as the spectral slope at the cutoff, completes χ near the crossover, and whether the map carries over to three dimensions.

Code and data availability. The solver, the analysis tools (`tools/compare_schemes.py`, `tools/cascade.py`, `tools/modified_wavenumber.py`) and the command lines of every run are in the cnavier repository (<https://github.com/leo-aa88/cnavier>), with a detailed methodology document (`docs/cnavier_methodology.pdf`).

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